```
PS C:\Users\molef> nslookup google.com
Server:  UnKnown
Address:  192.168.0.1

DNS request timed out.
    timeout was 2 seconds.
Non-authoritative answer:
Name:    google.com
Addresses:  2c0f:fb50:4002:801::200e
          142.251.47.46
```

```
PS C:\Users\molef> nslookup -type=MX gmail.com
Server:  UnKnown
Address:  192.168.0.1

DNS request timed out.
    timeout was 2 seconds.
Non-authoritative answer:
gmail.com       MX preference = 10, mail exchanger = alt1.gmail-smtp-in.l.google.com
gmail.com       MX preference = 5, mail exchanger = gmail-smtp-in.l.google.com
gmail.com       MX preference = 30, mail exchanger = alt3.gmail-smtp-in.l.google.com
gmail.com       MX preference = 40, mail exchanger = alt4.gmail-smtp-in.l.google.com
gmail.com       MX preference = 20, mail exchanger = alt2.gmail-smtp-in.l.google.com
```

```
C:\Users\molef>nslookup -type=MX github.com
Server:  UnKnown
Address:  192.168.0.1

Non-authoritative answer:
github.com      MX preference = 0, mail exchanger = github-com.mail.protection.outlook.com

github.com      nameserver = dns1.p08.nsone.net
github.com      nameserver = dns2.p08.nsone.net
github.com      nameserver = dns3.p08.nsone.net
github.com      nameserver = dns4.p08.nsone.net
github.com      nameserver = ns-1283.awsdns-32.org
github.com      nameserver = ns-1707.awsdns-21.co.uk
github.com      nameserver = ns-421.awsdns-52.com
github.com      nameserver = ns-520.awsdns-01.net

C:\Users\molef>
```

Resolving Query records manually can help us understand what A, CNAME, MX AND TXT records do. We can also Query public DNS Domain Name Severs directly bypassing the local resolver.

The IP Address that google.com query resolves to is **142.251.47.46**.

The MX record is a DNS record that directs email to a specific mail server responsible for receiving messages for a domain. It is responsible for email routing.

```
PS C:\Users\molef> nslookup github.com
Server:  UnKnown
Address:  192.168.0.1

DNS request timed out.
    timeout was 2 seconds.
Non-authoritative answer:
Name:    github.com
Address:  20.87.245.0
```

When querying an ISP's DNS we can use the nslookup command to see the router IP Address **192.168.0.1** and the DNS we are querying's address **20.87.245.0**. It simply shows us the source address and the web server address.

TTL Time to Live is a setting in seconds that's determines how long routers cache a DNS record before fetching an update from a server of authority. We can see in the shot above that the request timeout was 2 seconds for a request to the github.com server.